FLEET TEAM DIGEST

737MAX-FTD-32-17023

Issue Title : Tightening Method for Wheel Tie Bolts - 737 Carbon Brake Wheels

Airplane Model	ATA	Minor Model(s)
737MAX	3245-11	-7, -8, -8200, -9, -BBJ

Other Model(s)	ECCN
737NG	9E991

Originated Date	Last Revised Date	Created On
05/11/2017	04/19/2018	03/02/2018

Estimated Completion Date	Next Update Date

Revision Description

March 2018: Revised to indicate that Safran (Messier-Bugatti) has changed the wheel tie bolt torquing procedure in their CMM to require torque-angle methods.

Revised text to reflect current company names.

Applicability

Applies to all 737MAX airplanes using carbon brakes and associated wheels.

Description

The wheel designs utilized for 737 airplanes using carbon brakes are unique in that both inboard and outboard bearings are contained in one wheel half. Service experience shows that this wheel design is more sensitive to proper wheel tie bolt tensile preload than other wheel designs.

This article discusses improved tie bolt tightening methods that have been developed for these wheels.

Background

Service experience has shown that if the preload in the tie bolts is insufficient when the affected wheels are assembled, the nuts may start slowly unscrewing from the bolts as the wheel is operated in service.

The mechanism that causes the nuts to unscrew is not well understood and has been difficult to duplicate in the lab. However, it is believed that as the wheel rolls under load, a very small motion occurs between the two wheel halves. This motion results in the nuts gradually backing off of the bolts. If the preload in the bolts is low, then more motion can occur between the wheel halves, thus increasing the tendency of the nuts to unscrew.

Excessive motion between the wheel halves results in increased fretting on the wheel mating faces. However, in a few extreme cases, the nuts unscrewed from the bolts to the extent that the wheel halves flexed apart sufficiently to cause a wheel rupture. Refer to the referenced FTD article for examples of wheel ruptures caused by insufficient preload on tie bolts.

When torquing the tie bolts, the tensile preload can be below the desired level for a variety of reasons including insufficient lubrication on the bolts, incorrect lubrication products, or worn / damaged threads.

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Testing has shown that tightening the bolts with a procedure called "torque-angle" (sometimes called "snug-angle") results in more consistent preload in the tie bolts as compared to using a standard torquing method.

In the torque-angle process, the bolts are snugged up to a specified torque value, just enough to clamp the wheel halves together. Then instead of a final torque, the nut is turned through a specified angle to its final location.

Status

Closed.

Interim Action

None - see final action below.

Final Action

Both UTAS (Goodrich) and Safran (Messier-Bugatti) have developed torque-angle tightening procedures for the listed wheels.

The torque-angle method is now mandatory.

- UTAS has revised CMM 32-42-28 to require the torque-angle process for tie bolt torquing.

- Safran has revised CMM 32-49-83 to require the torque-angle process for tie bolt torquing.

Operator Action

We recommend that operators and/or their MRO's ensure that only the specified torque-angle method be used to tighten the wheel tie bolts.

Reference Categories

Reference Type	Reference Number	Revision Number	Reference Date
Fleet Team Digest Article (FTD)	737NG-FTD-32-12008		
Component Maintenance Manual (CMM)	32-42-28 (UTAS)		
Component Maintenance Manual (CMM)	32-49-83 (Safran)		

Parts List

Part Name	Boeing Part Number	Supplier Name	Supplier Number
Wheel, Main Landing Gear	S277A016-351	UTAS (Goodrich)	3-1674
Wheel, Main Landing Gear	S277A016-551	Safran (Messier-Bugatti)	C20626200

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Related Categories

- Component Issues : Issues that occur during component maintenance or have an impact on component reliability

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